

IGA Supermarket — Bill of Materials (BOM)

32-Channel Edge AI Surveillance | Base: Intel Core i5 | RX 9060 XT (16GB) | 16GB DDR5 | 512GB SSD + NAS Storage

Mandatory Materials

Component	Description / Specification	Qty
Dedicated AI GPU	AMD Radeon™ RX 9060 XT (16GB GDDR6, 128-bit) [Alt: RX 7600 XT 16GB]	1
Host Processor (CPU)	Intel Core i5-13400F or i5-14400F (10 Cores, 16 Threads, 65W TDP)	1
Motherboard	Intel B760M Micro-ATX (2.5GbE LAN, Dual M.2 PCIe 4.0)	1
System Memory (RAM)	16GB (2x 8GB) DDR5-5200 MHz CL36 Dual-Channel Kit	1
Local Storage (SSD)	512GB M.2 PCIe 4.0 NVMe SSD (WD Blue SN580 / Crucial P3 Plus)	1
Power Supply (PSU)	650W 80+ Bronze / Gold Certified ATX Power Supply	1
Computer Chassis	Micro-ATX High-Airflow Mesh Case with 4x Pre-installed Cooling Fans	1
CPU Cooler	Thermalright Assassin X 120 Refined SE (120mm Tower Cooler)	1
Secondary Network Card	PCIe 2.5 GbE Network Card (Realtek RTL8125B) for isolated CCTV VLAN	1
48-Port PoE+ Switch	TP-Link Omada TL-SG3452P (48x GbE PoE+, 384W Power Budget, 4x SFP)	1
Cat6 Patch Panel	1U 19" 48-Port Cat6 Unshielded Patch Panel (110 Punch-Down / Keystone)	1
Rackmount 8-Outlet PDU	1U 19" Horizontal PDU Power Strip with Surge Suppression	1
Cat6 Molded Patch Leads	0.5m & 1.0m Snagless Slim Cat6 Patch Cords (Switch to Panel)	35
RJ45 Connectors	Cat6 Pass-Through RJ45 8P8C Plugs + Strain Relief Boots	100 pk
Existing NAS Storage	Supermarket NAS mount via NFS/SMB for 24/7 video loop (24 MB/s stream)	Existing

Optional Materials (Site-Dependent / Upgrades)

Component	Description / Specification	Qty
Commercial 2.5GbE Router	TP-Link Omada ER707-M2 (2x 2.5GbE, 4x GbE, 802.1Q VLANs) — Needed if current router lacks 2.5GbE/VLANs	1
Battery Backup (UPS)	APC Smart-UPS SMT1500IC 1500VA/1000W Pure Sine Wave with USB cable — Needed if site lacks 1500VA UPS	1
Surveillance Cameras (32x)	32x Dahua 4MP WizSense IP Cameras (Aisles, Produce, Bakery, Checkouts with Mic, IK10 Domes) — If replacing	32
32GB RAM Upgrade	32GB (2x 16GB) DDR5-5600 CL30 — Upgrades base 16GB for 10GB RAM-disk pre-alarm buffer	1
Equipment Server Cabinet	9U or 12U Wall-Mount Network Enclosure (600mm Depth, Glass Door) — If rack space unavailable	1
Bulk Cat6 Network Cable	305m (1000ft) Box Solid Bare Copper Cat6 UTP 23AWG (2 boxes = 610m) — If re-cabling runs	2 boxes
Horizontal Cable Manager	1U 19" Finger Duct Cable Management Panel with Removable Cover	1
Cellular Failover Modem	4G/5G USB LTE Dongle (Plugs into router for automatic cellular internet failover)	1